From Research Plots to Prairie Fields: Do Ultra-early Seeding and Enhanced Efficiency Fertilizers Pay?

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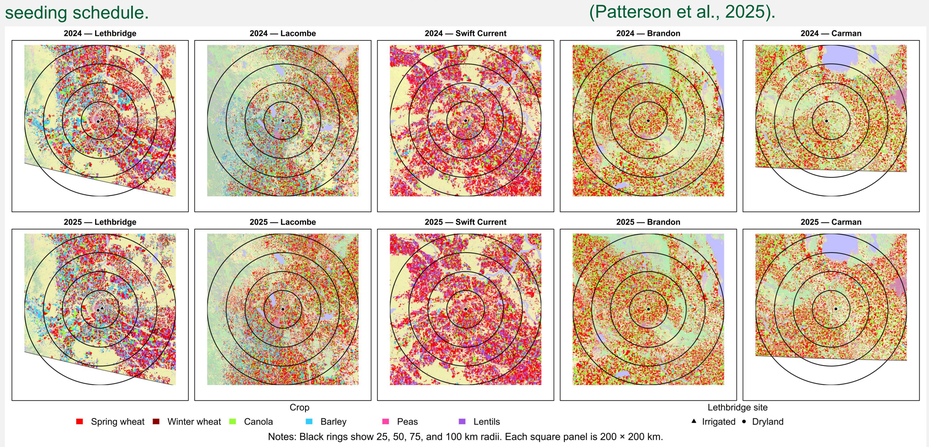
WHY SEED EARLIER OR CHANGE FERTILIZER?

Introduction

Farmers pay the costs of a new practice before knowing its yield benefit. Ultra-early seeding may improve spring wheat production, while enhanced-efficiency fertilizers (EEFs) may reduce nitrogen losses. Whether either practice pays depends on local growing conditions, fertilizer premiums and the cost of changing the farm's seeding schedule.

Background

Ultra-early seeding can improve wheat yield stability (Toerper et al., 2026). EEF yield benefits vary across environments, with some soil zones showing a marginal advantage over urea (Fast et al., 2023). Few studies have combined regional trials and local experiments to inform nitrogen decisions, although transferring that information can be difficult (Patterson et al., 2025).



Notes: Black rings show 25, 50, 75, and 100 km radii. Each square panel is 200 × 200 km.

The seeding dates for all field trial locations were approximately between seeding in April versus May, with Lethbridge being March versus April. The fertilizer products considered across the two seeding dates were urea, SuperU, and eNtrench.

Objectives

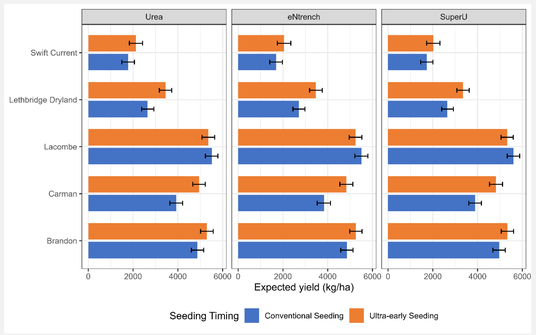
Compare how seeding date and fertilizer choice affect spring wheat yield, financial returns, and N_2O losses. Determine where these changes are more likely to pay using Saskatchewan rural municipalities (RMs) to train a machine learning (ML) model called BART.

Data

Field trial data is supplied by Agriculture and Agri-Food Canada (AAFC) for the 2024–2025 trials comparing two seeding dates and three fertilizers. Local data combined with Environment and Climate Change Canada weather station, AAFC 100m Soil Landscapes of Canada, Annual Crop Inventory satellite data, soil zone specific nitrogen rates, and Saskatchewan RM yields from crop insurance data are used to train the ML model.

YIELD RESPONSES

Ultra-early seeding increased spring wheat yields across fertilizer types, but the size of the gain varied by location. EEFs produced yields similar to untreated urea. Hence, their price premium are not offset by a yield gain.



Larger yield gains occurred at sites with higher growing degree days. Local conditions matter when applying trial results.

FINANCIAL RETURNS

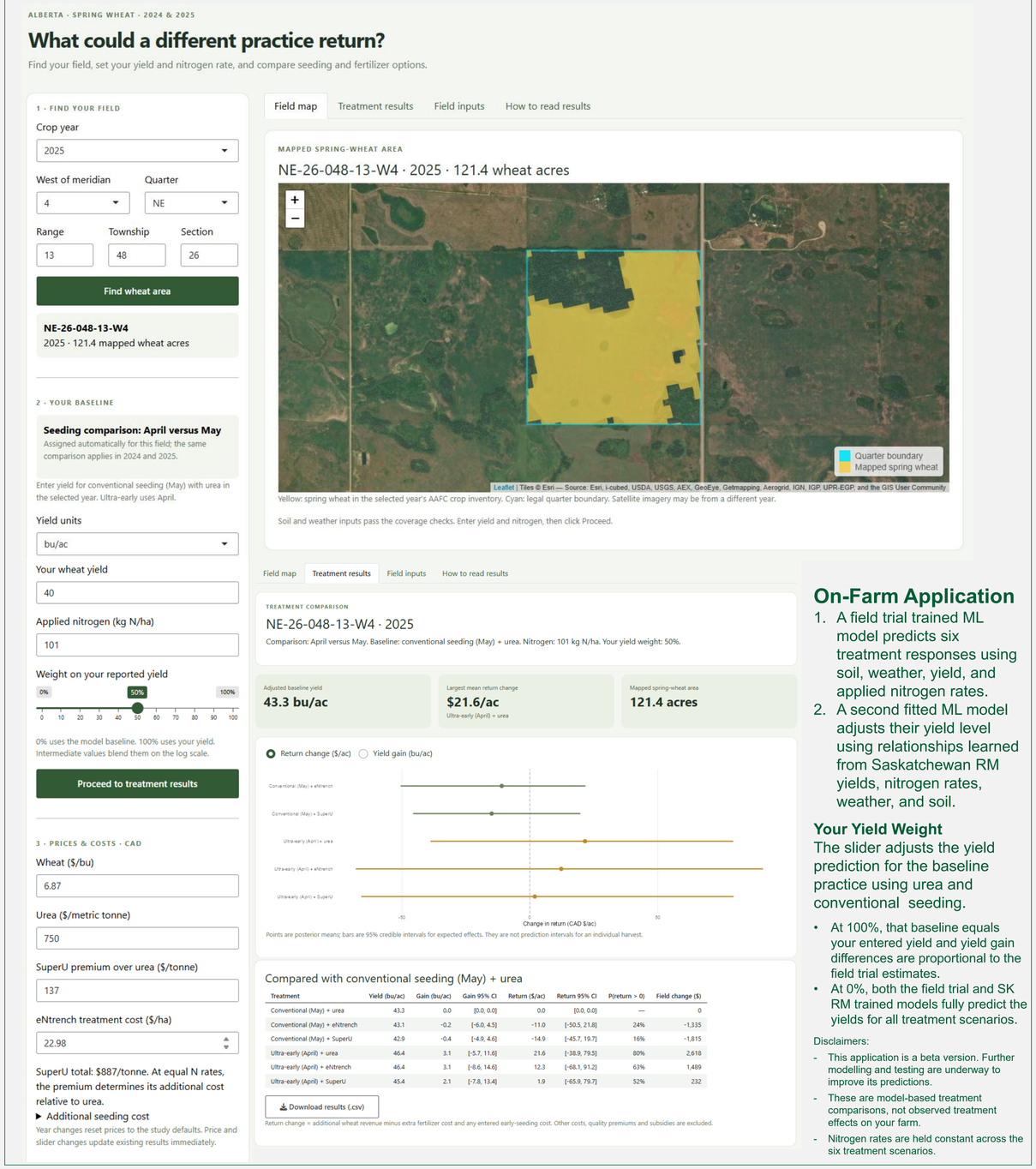
The increase in financial returns is the change in grain sales minus the change in fertilizer costs from seeding earlier rather than at the conventional date. Estimated returns by location and year are shown below, with totals for a 160-acre field in brackets:

Brandon, MB **2024: +$72/acre ($11,520/field)** **2025: +$28/acre ($4,480/field)**	**Lethbridge Dryland, AB** **2024: +$129/acre ($20,640/field)** **2025: +$59/acre ($9,440/field)**
Carman, MB **2024: +$222/acre ($35,520/field)** **2025: +$27/acre ($4,320/field)**	**Lethbridge Irrigated, AB** **2024: +$52/acre ($8,320/field)** **2025: -$72/acre (-$11,520/field)**
Lacombe, AB **2024: +$10/acre ($1,600/field)** **2025: -$40/acre (-$6,400/field)**	**Swift Current, SK** **2024: +$74/acre ($11,840/field)** **2025: +$11/acre ($1,760/field)**

Earlier seeding increased estimated returns at most locations, but the benefits were smaller in 2025, with losses at Lacombe and Lethbridge Irrigated. These results occurred during an exceptional year for crop production, with farmers harvesting a record spring wheat crop. Estimates do not deduct the cost of delaying another crop to seed spring wheat earlier

Prices: Saskatchewan Crop Planning Guides, Corteva, Manitoba Agriculture and Koch Agronomic Services.

WHAT COULD THIS MEAN FOR YOUR FARM?



On-Farm Application

1. A field trial trained ML model predicts six treatment responses using soil, weather, yield, and applied nitrogen rates.
2. A second fitted ML model adjusts their yield level using relationships learned from Saskatchewan RM yields, nitrogen rates, weather, and soil.

Your Yield Weight
The slider adjusts the yield prediction for the baseline practice using urea and conventional seeding.

- At 100%, that baseline equals your entered yield and yield gain differences are proportional to the field trial estimates.
- At 0%, both the field trial and SK RM trained models fully predict the yields for all treatment scenarios.

Disclaimers:

- This application is a beta version. Further modelling and testing are underway to improve its predictions.
- These are model-based treatment comparisons, not observed treatment effects on your farm.
- Nitrogen rates are held constant across the six treatment scenarios.

REFERENCES

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